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Wind energy penetration impact on active power flow in developing grids

Innocent O. Ozioko[a,*], Nnaemeka S. Ugwuanyi[b], Arthur O. Ekwue[c], Charles I. Odeh[a]

[a] Department of Electrical Engineering, University of Nigeria Nsukka, Nigeria
[b] Department of Electrical/Electronic Engineering, AE- FUNAI, P.M.B. 1010, Abakaliki 480214, Nigeria
[c] Department of Electrical Engineering/Africa Centre of Excellence on Sustainable Power & Energy Development, University of Nigeria, Nsukka, Nigeria

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A B S T R A C T

Renewable energy (RE) generators are typically connected to the grid via the distribution network or transmission network load buses. With today's large-capacity RE generators, such as wind turbines, and the environmental concerns, these RE generators could become the grid's major power source. Thus, renewable energies such as wind are predicted to gradually replace traditional generators. This raises questions about which generator to be replaced first, and what the acceptable limit of RE penetration will be? In this paper, the impact of wind energy penetration on the active power flow in the Nigerian grid was investigated. The integration via load buses was considered, as well as the integration by gradually replacing conventional generators with DFIG-based wind turbine generators (WTG). In each case, the maximum wind energy penetration was determined. The results show that the penetration of wind energy is more when connected to load buses than when it replaces existing generators. Furthermore, when replacing a Nigerian gas-fired generation with offshore WTG, Bus 2 generator (Delta plant) is the first contender, which minimizes the active power loss. However, if the WTG will link via a PQ bus, Bus 11 (Aja) is the best connection point.

Introduction

One of the primary issues confronting developing countries is a deficiency of energy supply that can keep up with rising demand. Most developing countries like Nigeria depend largely on non-renewable energy sources that diminish quickly [1]. For example, about 80% of generation in Nigeria comes from thermal plants, while the remaining supply is from hydro plants [2]. Many homes and companies rely on fossil fuel generators for power supply alternatives [3], which significantly increase the greenhouse gases. These techno-economic challenges contribute to the rising interest in renewable energies

* Corresponding author.
E-mail addresses: innocent.ozioko.pg01268@unn.edu.ng (I.O. Ozioko), ugwuanyinnaemeka@gmail.com (N.S. Ugwuanyi), arthur.ekwue@unn.edu.ng (A.O. Ekwue), charles.odeh@unn.edu.ng (C.I. Odeh).

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Nomenclature

WTG	Wind Turbine Generator
DFIG	Doubly Fed Induction Generator
REs	Renewable Energies
GFG	Gas Fired Generator
P_{therm}	Thermal plants' equivalent real power.
P_{hydro}	Hydro plants' equivalent real power.
$PG_{i_{new}}$	i-th generator's new capacity after wind penetration
P_{loss}	Real power loss
PG_i	i-th conventional generators capacity
X	Line reactance.
I_i	i-th Line Current
R_i	i-th Line Resistance
P_i	Real power flow on the i-th line
Q_i	Reactive power flow on the i-th line
V_i	i-th line's voltage drop.
P_w	Wind turbine's real power.
p.u	Per Unit
ng	Number of generators.

(REs) globally. Nigeria has abundant potentials of REs such as Hydro, Wind, Solar PV, Biomass, etc.[4–6], especially the wind energy [7,8] (both in offshore and onshore [9,10]), which can be tapped for sustainable energy generation. Sadly, there is currently no wind farm interconnected to the Nigerian grid, though a 10 MW onshore project is under development. The technology development of wind energy gives it an edge over other REs with several advantages like easier installation when compared with a hydro turbine, better available location, etc. [11]. Offshore wind generation in recent times is developing faster than onshore due to steady, smooth, and high wind speed over onshore globally. REs are often connected near load centers, where they are referred to as distributed generation (DG).

The integration of wind energy into the grid, like other REs, may have an impact on system integrity, either favorably or adversely. Connecting wind power to the grid can cause power quality concerns such as voltage profile variations, harmonics and grid stability [12–14].

Power flow is bidirectional with DG connection, hence, any DG (both conventional and renewable generators) can change the voltage profile in a network. Several researches [15–20] support that optimally sited DG in a power system improves the voltage profile and reduces the active power losses.

Optimization strategies are employed to ensure that the wind farm is sited optimally [21,22]. In fact, authors in [23] and [24] considered power loss minimization in a network as a factor in determining the proper operation of wind turbines embedded in a power system. Furthermore, the improvement of small-signal stability [25,26] and transient stability [27] of power systems with the integration of DG such as wind energy has been reported in the literature.

Although the optimal wind farm integration can improve the bus voltage, the stability, and reduce the power loss of the system, the improvement depends on the penetration level as well as the location of a wind turbine [28,29]. There seems to exist no exact penetration limit for wind energy. For example, in Denmark, about 20% penetration limit was reported in [30]. Some researchers consider the penetration level from 20 – 30% as high penetration, and 1 – 5% as low penetration [28].

The bulk of the above studies concentrated on wind energy integration into the grid via either a load bus or a distribution network. Moreover, the effect of replacing the existing conventional generator with wind turbine was not considered in the analyses. As wind energy penetration via PV buses was not examined, findings on the penetration limit, power loss reduction, voltage profile improvement, e.t.c., are not general. In the Nigerian power system context, most of the researchers interests are typically in the use of REs to increase the available power. For instance, authors in [31–35] focused on either the prospects of wind energy or the quantification of harvestable wind energy, considering the specificity of the Nigerian annual wind speed. None of these studies accounted for the penetration limit of RE integration into Nigerian grid. Reference [20] considered the wind power penetration limit to maintain an acceptable voltage profile of the Nigerian power system. However, only the integration via a load bus was considered. Moreover, the conventional generators were not replaced with the WTG. O. Bamisile *et. al*[3] examined the maximum penetrable PV and wind energy in the Nigerian grid to match the present electricity demand without producing a critical surplus of electricity. Results indicate solar PV is less expensive than wind turbines for this purpose. However, technical aspects like power losses, voltage profile, and the implications of replacing existing fossil fuel generators with PV or WTG were not considered.

With the advancement of wind energy technology, it is clear that several conventional generators will be phased out in favor of WTGs. In this case, the WTG will be connected on PV buses, as opposed to the usual connection on load (PQ) buses. As this replacement will be a gradual process, the question of priority arises. Which one should be replaced first? Nowadays, the energy generated by an offshore wind farm may be several times more than that produced by a conventional generator. This poses a second question: what level of penetration is permissible after replacing the conventional generators?

The remainder of this paper is organized as follows: Section 2 presents background on wind energy integration and how to calculate the renewable share of total power. In Section 3, the simulation models together with the modelling parameters of the two power systems case studies employed in this study are discussed. In Section 4, the findings are presented and discussed. Section 5 concludes the article and presents possible future research.

Wind energy integration background

Renewable energies like the wind are predicted to replace conventional generators in the future [36] due to the low carbon emission and regenerative nature of this kind of energy resource. As noted earlier, the replacement may not happen all at once, but rather in stages. As a result, it is important to look into replacing fossil fuel driven generators in the network with a renewable energy generator (like DFIG-based WTG as in our case), especially in the Nigerian power system where the main generation is via fossil fuel. Thus, power flow models should accurately account for the WTG.

Power flow models with wind turbine

This subsection briefly highlights the possible models for incorporating DFIG-based wind energy conversion system (WECS) in load flow analysis (LA). Several studies have presented different possibilities for incorporating WTG in a power flow analysis. Examples include the PQ, PV, and RX models [37]. The power flow model adopted for incorporating DFIG-based WECS in a power system mainly depends on the control objective. Authors in [38] opined that the PV model is convenient when voltage is controlled, while the PQ model can be preferred when the power factor is controlled. For the details of PQ and RX models, the reader can refer to [37]. The PQ model is more common in literature and is adopted in this research. Moreover, it allows controlling the power factor to focus strictly on the active power flow, which is our main focus.

Related works

In this section, some works related to this research are summarized. The related works are summarized in Table 1.

Table 1
Summary of the related works.

Citations	Year	Approach	Merits	Limitation
[24]	2007	Impact of location and sizing of DG studied.	Real and reactive power improvement	Only penetration via PQ bus considered. No penetration limit.
[15]	2014	Synchronous generator replaced with a WTG	Power quality improvement.	No penetration limit.
[16]	2015	Benefits of DGs investigated.	Technical loss/voltage improvement.	Penetration via PQ bus No penetration limit.
[29]	2016	Power quality investigation with WTG as a DG.	Power quality improvement	Only penetration via PQ bus considered. No penetration limit.
[18]	2016	DG's impact on Nigerian grid studied.	Power loss/quality improvement	Only PQ buses considered without penetration limit.
[3]	2020	Matching demand with PV and wind integration	PV and Wind penetrable limits determined	Technical aspects and genertor replacement not considered.
[12]	2021	Effect of injecting wind power to the grid studied.	Voltage Improvement.	Only injection via PQ buses considered. No penetration limit.
[19]	2021	DGs impact on protection scheme studied	Power quality improvement. Increased fault current observed.	Varying degree of DG missing. No Penetration limit.
[17]	2021	DG's impact on Nigerian grid studied.	Power loss/quality improvement	Only PQ buses considered without penetration limit.
[20]	2021	DFIG's impact on the voltage profile of the Nigerian grid	Bus voltage improvement	Only load buses considered. Real power loss missing.
[22]	2021	Optimal sizing and location of DGs	Power quality enhancement	PQ buses considered. No penetration limit.

Wind energy penetration

In any power system, the equivalent real power loss can be obtained from the calculation of individual line loss in the system. For n lines, the total real power loss can be expressed as [28]:

$$P_{\text{loss}} = \sum_{i=1}^{n} |I_i|^2 \cdot R_i = \sum_{i=1}^{n} \left(\frac{P_i + Q_i}{V_i} \right)^2 \cdot R_i, \tag{1}$$

where I_i is i-th line current, R_i is i-th line resistance, P_i is the active power flow on the i-th line, Q_i is the reactive power flow on the i-th line, and V_i is the voltage drop on the i-th line.

Several wind turbines are usually characterized as real power generators [15,28]. If the WTG generates only real power, then, the power is injected into the grid at a unity power factor. Hence, equation (1) can be simplified as

$$P_{\text{loss}} = \sum_{i=1}^{n} \left(\frac{P_i}{V_i} \right)^2 \cdot R_i. \tag{2}$$

It should be noted that WTG does generate reactive power, but this simplification is necessary and suitable to study the real power flow. Therefore, the percentage share of wind power in terms of real power can be expressed as [28]:

$$\%P_w = \frac{P_w}{\sum \left(P_{G_i} + P_w \right)}, \tag{3}$$

with $\%P_w$, percentage wind energy penetration, P_w, the capacity of wind energy penetration in p.u., and P_{G_i}, i-th conventional generators capacity in p.u. If the system already has certain renewable energy sources, such as hydropower, as is the case with the Nigerian grid, equation (3) can be modified as follows:

$$\%P_w = \frac{P_w}{\sum \left(P_{therm} + P_{hydro} + P_w \right)}, \tag{4}$$

with P_{therm}, equivalent real power for thermal plants in p.u., P_{hydro}, equivalent real power for the hydro plants in p.u. To maintain the same generation, the new capacities of each of the conventional generators following the wind energy integration becomes:

$$P_{Gi_{new}} = \frac{P_{G_i}}{\sum_{i=1}^{n_g} P_{G_i}} \left(|P_w - \sum_{i=1}^{n_g} P_{G_i}| \right), \tag{5}$$

with $P_{Gi_{new}}$, the new capacity of the generator i at any wind penetration in p.u., P_w, wind energy penetration/capacity in p.u., and n_g, number of the considered generator.

Simulation models

Two power system case studies were used for this investigation: the IEEE 5-bus power system, and the Nigerian power systems. In this section the simulation models and parameters of the case studies are briefly described. The parameters are given in p.u. with 100 MVA base.

Case study 1: IEEE 5-bus system

The single-line diagram of the IEEE 5-bus power system is shown in Fig. 1 with slack at bus 1.

The network contains two synchronous machines connected at buses 1, and 2 respectively, with three connected loads. It contains seven transmission lines of different impedance parameters. The network has a total base case generation of 1.486 p.u. and a total load of 1.45 p.u. This network has been used for similar studies in [28] and is used again here to allow result comparison. Although simple, it reveals the main phenomena of interest in this research. The parameters for the model can be found in Appendix A.

Case study 2: 28-bus nigerian power system

Fig. 2 shows the single-line diagram of the 330kV, 28-bus Nigerian power system containing 9 synchronous generators, 24 loads, and 53 feeders. Each of the feeders has an RL series impedance.

The modeling parameters were drawn from [39] and are presented in Appendix B. The base case generation is 43.786 p.u., with a total load of 42.718 p.u on a 100 MVA base power. We have neglected variations in generation and demand for the sake of simplicity. In practice, the impact of load-shedding and constrained generation is taken into account. This network comprises two generator types, namely, the hydro plants and the gass-fired generators (GFGs) located in the northern and southern parts of the country, respectively. Almost all the GFGs are located in the southern part of the country, making

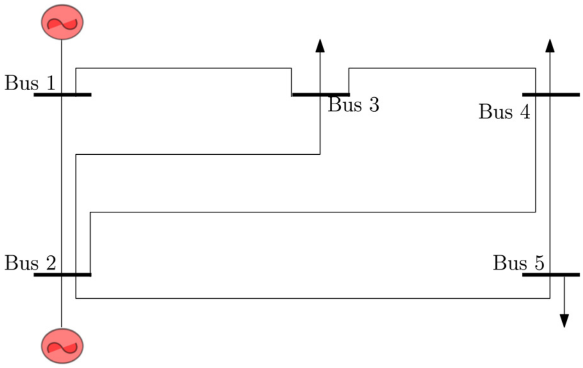

Fig. 1. IEEE 5-bus network.

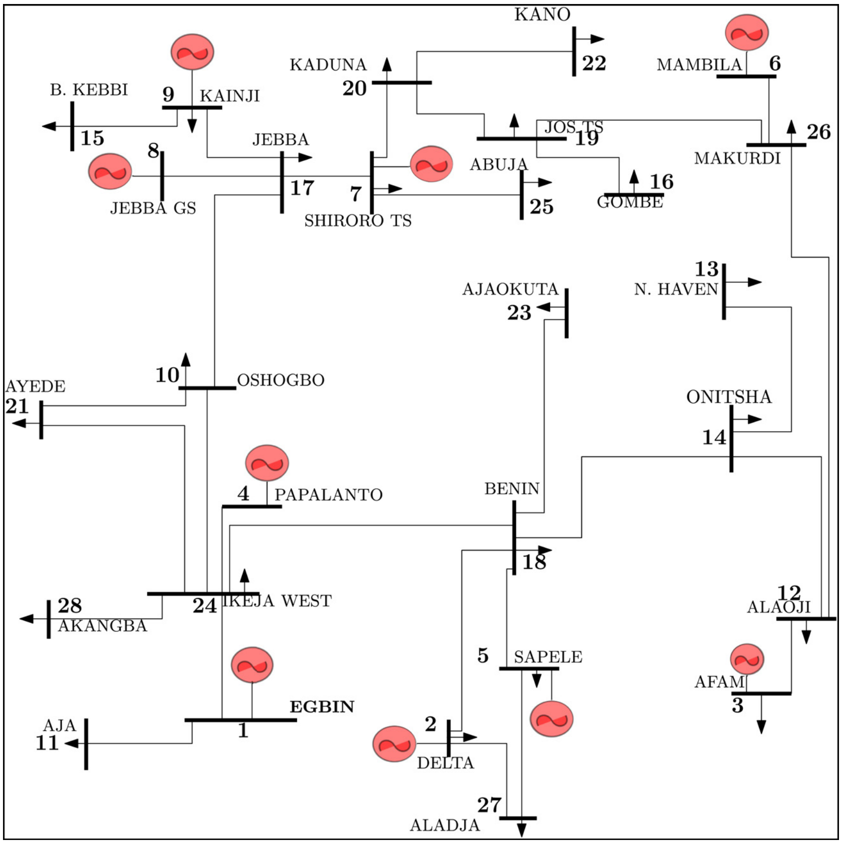

Fig. 2. The 28-bus Nigerian power system with WTG at bus 2.

the gas emission in the region enormous. The network can benefit from offshore wind energy, as already noted by [10]. Moreover, the southern part of Nigeria is bounded by the Atlantic Ocean. As a result, some gas-fired generators may be replaced by offshore wind farms.

The two networks (Figs. 1 and 2) were modelled in PSAT® software in MATLAB® environment. The DFIG-WTG was modelled as a PQ load with a negative generation at a unity power factor. The model paramters for the DFIG-WTG are listed in Table C.6. The voltage limit of ±5% from 1 p.u. was considered. We assumed the wind speed remains constant. Wind variability is different in different places, but maybe averagely the same in a particular place. Moreover, considering the active power flow by a steady-state load flow solution, the assumption does not mar the expected phenomena, notably, the power losses.

Results and discussions

In this section, the simulation results for the two case studies are presented and discussed. In each case, the base case conditions of the system were obtained through Newton-Raphson load flow analysis before the wind integration.

Case study 1: IEEE 5-bus system

The single-line diagram of the IEEE 5-bus power system has been presented in Fig. 1. Firstly, the conventional generators were successively replaced, and the power flow ran to determine the total real power loss. In other words, the WTG was connected to PV buses. Secondly, the WTG was connected to PQ buses, but conventional generators' capacity was adjusted, to keep the total system generation constant.

Conventional generator replacement with DFIG-WTG

Fig. 3 a presents the real power loss with the WTG sequentially connected to buses 1 and 2. The base case (i.e., without WTG) result is also plotted on the figure for comparison. The loss is minimal when the WTG is connected to bus 1 and is approximately the same as the base case. When the WTG is placed on bus 2, the real power loss is more than twice the base case. From the results, bus 1 provides the optimal replacement generator considering the power loss. Although the voltage is not controlled when the DFIG-based is modelled as a PQ load, the voltage profile should be monitored to ensure it does not deteriorate more than the base case. The corresponding voltage profile is shown in Fig. 3b. The Figure shows that the voltage limit was not violated irrespective of the generator replaced, but the replacement of the generator at bus 1 with a WTG provides a better voltage profile.

Although WTGs of the same capacity were utilized to replace conventional generators, the effects on real power and even voltage profile varied with the generators. As a result, the issue of precedence during replacement becomes critical.

DFIG-Based wind turbine generator connected to PQ bus

Authors noted in [15] that wind turbine is connected preferably at the strongest bus. The base case voltage profile of the network indicates the three PQ buses (Buses 3, 4, and 5) are similarly strong, having nearly the same voltage level. Buses 3, 4 and 5 have voltage levels of approximately 0.979 p.u., 0.979 p.u., and 0.977 p.u., respectively. As a result, any of them is viable for WTG connection. A 30% integration of WTG on the PQ buses 3, 4, and 5 recorded voltage levels of about 0. 993 p.u, 0. 993p.u, and 1.006p.u. as seen from Fig. 4a. The results indicate that the voltage limit is respected irrespective of the buses. The voltage profile was improved, which is consistent with earlier investigations in which the DG is connected to the load buses. The real power losses are shown in Fig. 4b and point that bus 5 is the optimum location of the WTG. However, we shall see later that the decision of optimum location depends also on the level of integration considered.

Determining the maximum wind energy penetration

To investigate the integration level, the WTG was initialized at 0.05 p.u. step up to certain percent penetration using equation (3). The three load buses were investigated and the results are shown in Fig. 5a. In all cases, the losses are less than the base case loss. Usually, the maximum penetration is considered as the point where the loss is minimum. It can be seen from the figure that when the WTG is located on bus 5, the minimum loss is at 57% wind penetration. Therefore, the limit is 57% on bus 5. Beyond this limit the losses starts increasing. The authors in [28] carried out a similar study on the same system and identified bus-5 as the best connection point for the WTG at 30% total generation, which is coherent with Fig. 5a.

It should be noted that the maximum penetration depends on the location of WTG. For instance, buses 3 and 4 seem to allow more penetration, if other constraints like the voltage profile are ignored since their losses are lesser than that of bus

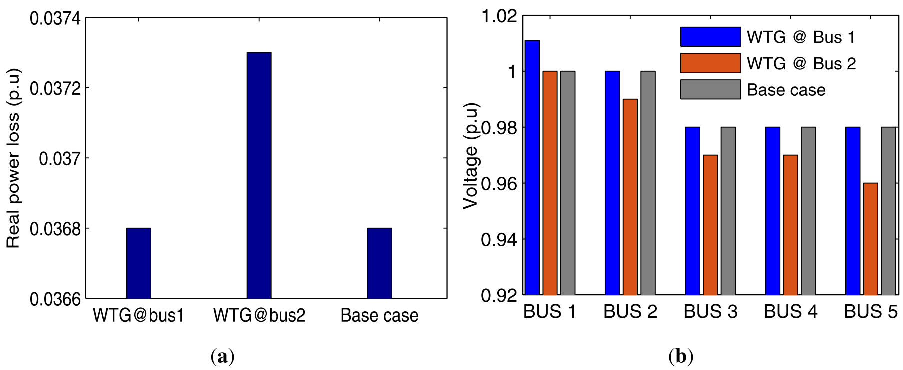

Fig. 3. (a) Real power loss for IEEE 5-bus system with WTG on PV buses. (b) Voltage profile of the IEEE 5-bus system with WTG on PV buses.

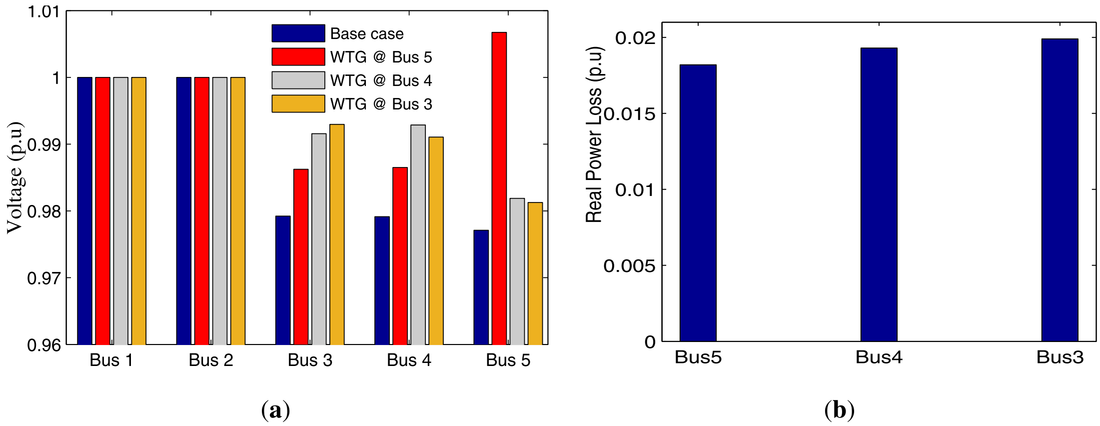

Fig. 4. (a) Voltage profile of the IEEE 5-bus system with WTG on PQ buses. (b) Real power loss of the IEEE 5-bus system with WTG on PQ buses.

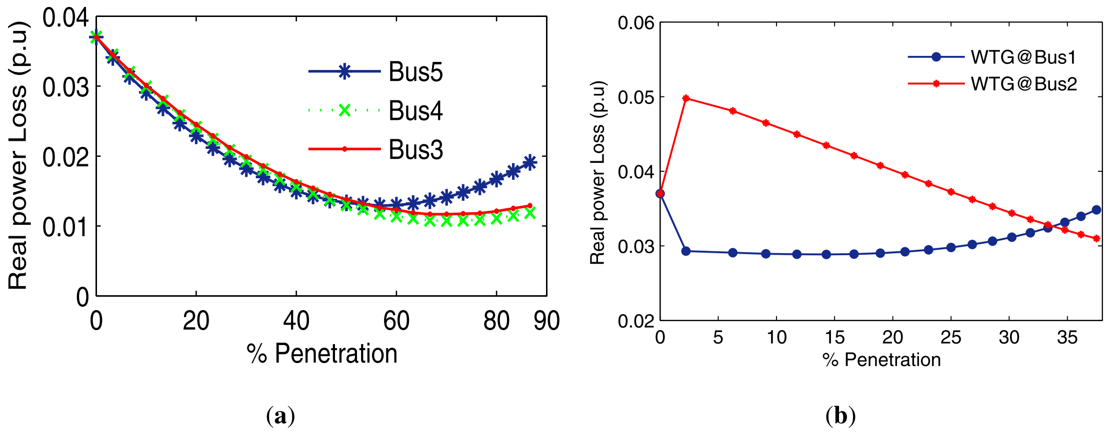

Fig. 5. (a) Real power losses against %wind penetration for IEEE-5bus system at PQ buses. (b) Real power losses against %wind penetration for IEEE-5bus system at PV buses.

5 beyond 57% penetration. This observation indicates the optimum location depends on size of the WTG. For example bus 5 is the optimum location of WTG for penetration between 0 and 57%, while buses 3 and 4 are better beyond this penetration provided other constraints are respected. The penetration level was equally investigated for the case where the generators are replaced with a DFIG-WTG as shown in Fig. 5a. Generator at bus 1 had already been identified as the best replacement generator following Fig. 3. The minimum loss for the integration at bus 1 is at about 14%, which is far lesser than 57% found for penetration via PQ buses.

The following deductions can be made from the observations above: i) The penetration is more when the WTG is connected to a load bus than when used to replace an existing conventional generator; ii) While the losses are still below the base case, the level of integration must be factored in determining the best connection point. Although the above observations are clear, owing to the loading and structure of the IEEE 5-bus system, the differences in values are not very obvious. In the next section, the Nigerian power system will be investigated similarly,.

Case study 2: 28-bus nigerian power system

The single-line diagram of the Nigerian 28-bus power system has been presented in Fig. 2. Currently, RE generators are in the North while the GFGs are in the south. Reference [9] has shown that the north may be a viable option for onshore WTG integration because of an excellent wind speed available. However, to investigate the conventional generator replacement, we focus only on the southern part of the system since the generator types in the north are already renewable (hydro). As noted earlier, the southern part of the network is closer to the Atlantic Ocean, which is a potential location for offshore wind farm. Other researchers have also demonstrated the viability of offshore wind in the southern parts of Nigeria [8,10]. Therefore, our interest will be mostly in the integration of WTG at the south to reduce greenhouse emissions from the GFGs. Integrating WTGs into a grid system requires a lot of considerations. In the context of the Nigerian power system, many factors for selecting the buses to study are very important. The active power losses as well as the voltage profile of

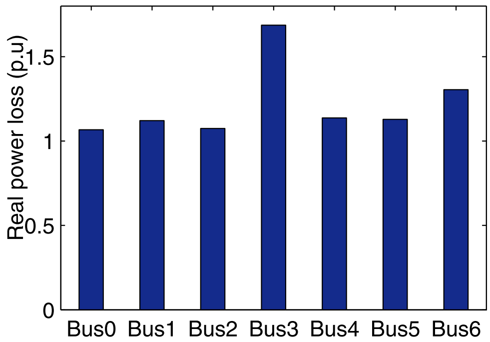

Fig. 6. Real Power Losses for integration of WTG into the Nigerian grid on PV buses.

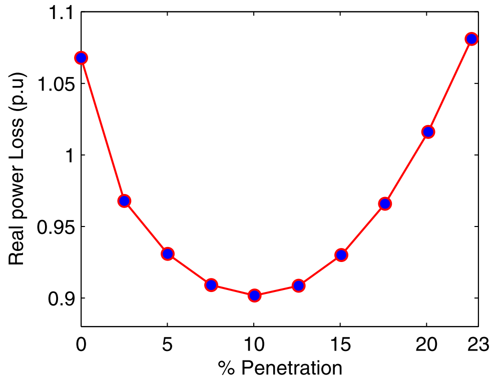

Fig. 7. Percentage wind power penetration into the Nigeria grid at bus 2.

the system from the load flow results as already demonstrated in the preceding section, the location of the offshore farm site, and the viable places for offshore wind generation were considered during the integration of WTG on both the PV buses and PQ buses.

Conventional generator replacement and integration limit

For PV bus WTG integration, all the generators excluding the hydro generators were replaced one after the other to identify the best replacement generator based on the load flow results (the power losses and voltage strength of the individual buses) in the Nigerian system. Technically, the PV bus that maintained the permissible voltage limit with the minimum power loss will be selected upon which the %Pw penetration will be examined. The wind power penetration was increased by 1.1 pu step, while the GFGs capacities were adjusted according to (5). Fig. 6 shows the real power loss after successively replacing the six GFGs with a DFIG-WTG of the same capacity. In the figure, Bus 0 corresponds to the base case. The loss is maximum when the generator at bus 3 is replaced, whereas replacing the generator at bus 2 gives the minimum power loss compared with all other buses. Thus, bus 2 is the optimum replacement generator.

The voltage profile (not plotted) confirms that replacing the generator at bus 2 gives the same voltage profile as the base case whereas the voltage profile deteriorates with other options. Fortunately, generator 2, located in Delta, is 4431km closer to the Bight of Biafra [40], a potential offshore wind farm site, than generators 4 and 1, which come next in terms of real power loss in Fig. 6. With bus 2 selected, the maximum penetration of the wind energy is determined next. The percentage penetration of wind power (%Pw) on bus 2 is presented in Fig. 7.

From the figure, the real power loss is minimum when the percentage integration is 10%, which implies that the maximum penetration is 10%. The 10% of 43.786 p.u. base case generation represents 4.4 p.u. wind power capacity.

DFIG-Based wind turbine generator connected to PQ bus and integration limit

As stated earlier, our attention is on offshore wind integration. So, only the load buses in the southern section of the network are considered. Among them, the strong buses were sorted. Finally, out of the strong buses, buses 11, 12 and 27 were closer to the Bight of Biafra, which has been identified as a potential offshore wind farm site. Therefore, only the investigations involving these three buses are reported. The percentage of wind energy penetration on these PQ buses were calculated and recorded. The result at a varying capacity of the WTG integration is presented in Fig. 8.

It can be observed that for the WTG at Bus 11 (Aja bus), the real power loss decreased with the percentage wind energy penetration until at 16% integration. Beyond this level of integration, the real power loss started to increase. Hence, the maximum penetration is 16%, which translates to 7 p.u. It can be observed that the placement of the WTG on buses 12

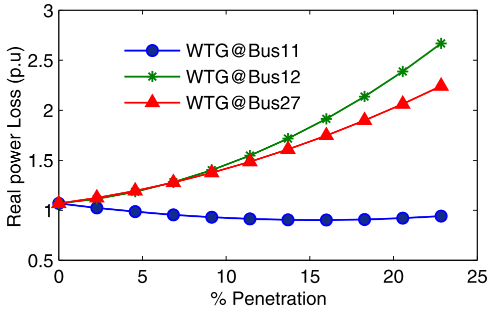

Fig. 8. Real power losses against %wind penetration at selected PQ buses for the Nigerian grid.

and 27 leads to an increasing real power loss, so they are bad choices. The voltage profile was also investigated (figures not shown) and we observed that all the options kept the voltage profile very close to the base case although the result for bus 11 gave the best profile.

From the results presented in all in Section, it can be observed that different networks and buses absorb power differently. For WTG connected to PQ buses, the IEEE 5-bus system has maximum wind integration of about 57% (i.e; at bus 5) whereas the Nigerian grid has a limit of about 16%. For the WTG integration via PV buses, the IEEE 5 bus power system has a limit of about 14% whereas the Nigerian grid has about 10%.

Conclusions

Proper sizing and placement of WTG improve the power system especially reducing power losses. In this paper, the effect of DFIG-based wind farm integration on the real power flow has been investigated. In addition, the permissible level of wind energy integration was explored. The study has been done with two different case studies: the IEEE 5-bus and the Nigerian power systems. From the results presented, the following conclusions can be drawn: (i) the DFIG-based wind energy penetration can increase or decrease the power losses in the system depending on the location; (ii) the connection point that gives optimal performance for small percentage wind energy penetration may not be the optimal connection point when the penetration level is increased, so there should be a balance between the penetration level and the location; (iii) more wind energy is integrated to the power system when connected via load bus than when replacing existing conventional generator if the wind generator generates only active power; (iv) when considering replacing the Nigerian gas-fired generator with the offshore wind generator, the first candidate to minimize the real power loss is the bus 2 generator (i.e. Delta plants). If the WTG will connect to the load buses, bus 11 (Aja bus) may be the first choice to reduce the real power loss; (v) the maximum percentage of wind energy penetration into the Nigerian power system is about 16% if the minimal active power loss is of interest. The observations from this work will serve to guide the operators in harnessing the available offshore wind energy, particularly in Nigeria and other developed countries at large.

In this work, we have considered a constant wind speed in the power flow analysis. In the future, wind speed variability will be accounted for in the power flow analysis. We have assumed the WTG integrated at a unity power factor with no voltage control, which is not ideal. It is also worth noting that the reactive power limits of conventional generators were not taken into account. This may have an impact on the results and merits further investigation in future work. In future, the WTG generating both reactive and active power should be considered in the analysis. Moreover, the impacts of DFIG-based WTG integration on the small-signal stability of the Nigerian grid should be investigated in future. Finally, in the future, economic studies for wind farm integration will be conducted.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Modeling Parameters for Case study 1: IEEE 5-bus system

Table A.2
IEEE 5bus line data.

Lines	R (p.u)	X (p.u)
1 - 2	0.015	0.06
1 - 3	0.079	0.24
2 - 3	0.058	0.18
2 - 4	0.055	0.18
2 - 5	0.035	0.12
3 - 4	0.005	0.03
4 - 5	0.075	0.24

Table A.3
IEEE-5bus load and generation data.

Load (p.u)		Generation (p.u)		Buses
P(p.u)	Q(p.u)	P(p.u)	Q(p.u)	
0.0	0.0	1.0	0.0	1
0.0	0.0	0.5	0.0	2
0.45	0.0	0.0	0.0	3
0.40	0.0	0.0	0.0	4
0.60	0.0	0.0	0.0	5

Appendix B. Modeling Parameters for Case study 2: 28-bus Nigerian power system

Table B.4
Line data of 28-bus Nigerian power system.

S/N	From bus	To bus	R(pu)	X(pu)	S/N	From bus	To bus	R(pu)	X(pu)
1	1	11	0.0006	0.0044	28	3	12	0.0010	0.0074
2	1	11	0.0006	0.0044	29	12	14	0.0060	0.0455
3	1	11	0.0006	0.0044	30	13	14	0.0036	0.0272
4	28	24	0.0007	0.0050	31	13	14	0.0036	0.0272
5	28	24	0.0007	0.0050	32	16	19	0.0118	0.0887
6	1	24	0.0023	0.0176	33	17	8	0.0002	0.0020
7	1	24	0.0023	0.0176	34	17	8	0.0002	0.0020
8	24	18	0.0110	0.0828	35	17	6	0.0096	0.0721
9	24	18	0.0110	0.0828	36	17	6	0.0096	0.0721
10	24	21	0.0054	0.0405	37	17	9	0.0032	0.0239
11	24	10	0.0099	0.0745	38	17	9	0.0032	0.0239
12	23	18	0.0077	0.0576	39	19	20	0.0081	0.0609
13	23	18	0.0077	0.0576	40	20	22	0.0090	0.0680
14	2	18	0.0043	0.0317	41	20	22	0.0090	0.0680
15	2	27	0.0012	0.0089	42	20	6	0.0038	0.0284
16	27	5	0.0025	0.0186	43	20	6	0.0038	0.0284
17	18	14	0.0054	0.0405	44	6	25	0.0038	0.0284
18	18	10	0.0098	0.0742	45	6	25	0.0038	0.0284
19	18	5	0.0020	0.0148	46	12	26	0.0071	0.0532
20	18	5	0.0020	0.0148	47	12	26	0.0071	0.0532
21	21	10	0.0045	0.0340	48	19	26	0.0059	0.0443
22	15	9	0.0122	0.0916	49	19	26	0.0059	0.0443
23	15	9	0.0122	0.0916	50	26	7	0.0079	0.0591
24	10	17	0.0061	0.0461	51	26	7	0.0079	0.0591
25	10	17	0.0061	0.0461	52	24	4	0.0016	0.0118
26	10	17	0.0061	0.0461	53	24	4	0.0016	0.0118
27	3	12	0.0010	0.0074					

Table B.5
Bus data of 28-bus Nigerian power system.

				Load		Generation			
Bus	Name	Type	Voltage (pu)	P(pu)	Q(pu)	P(pu)	Q(pu)	Qmin (MVar)	Qmax (MVar)
1	Egbin	Slack	1.05	0.689	0.517	5.13	0.00	-1006	1006
2	Delta	PV	1.05	0.00	0.00	6.70	0.00	-1030	1000
3	Afam	PV	1.05	0.525	0.394	4.31	0.00	-1000	1000
4	Papalanto	PV	1.05	0.00	0.00	7.5	0.00	-1010	1010
5	Sapele	PV	1.05	0.206	0.154	1.903	0.00	-1010	1010
6	Mambila	PV	1.05	0.00	0.00	7.5	0.00	-1010	1010
7	Shiroro	PV	1.05	0.703	0.361	3.889	0.00	-1010	1010
8	Jebba GS	PV	1.05	0.00	0.00	4.95	0.00	-1050	1050
9	Kainji	PV	1.05	0.07	0.052	6.247	0.00	-1010	1010
10	Osogbo	PQ	1.00	2.012	1.509	0.00	0.00	-	-
11	Aja	PQ	1.00	2.744	2.058	0.00	0.00	-	-
12	Alaoji	PQ	1.00	4.27	3.202	0.00	0.00	-	-
13	N/Haven	PQ	1.00	1.779	1.334	0.00	0.00	-	-
14	Onitsha	PQ	1.00	1.846	1.384	0.00	0.00	-	-
15	B/Kebbi	PQ	1.00	1.145	0.859	0.00	0.00	-	-
16	Gombe	PQ	1.00	1.306	0.979	0.00	0.00	-	-
17	Jebba	PQ	1.00	0.11	0.082	0.00	0.00	-	-
18	Benin	PQ	1.00	3.833	2.875	0.00	0.00	-	-
19	Jos	PQ	1.00	0.703	0.527	0.00	0.00	-	-
20	Kaduna	PQ	1.00	1.93	1.447	0.00	0.00	-	-
21	Ayede	PQ	1.00	2.758	2.068	0.00	0.00	-	-
22	Kano	PQ	1.00	2.206	1.429	0.00	0.00	-	-
23	Ajaokuta	PQ	1.00	0.138	0.103	0.00	0.00	-	-
24	Ikeja-west	PQ	1.00	6.332	4.749	0.00	0.00	-	-
25	Abuja	PQ	1.00	1.10	0.89	0.00	0.00	-	-
26	Makurdi	PQ	1.00	2.901	1.45	0.00	0.00	-	-
27	Aladja	PQ	1.00	0.965	0.724	0.00	0.00	-	-
28	Akamgba	PQ	1.00	2.447	2.058	0.00	0.00	-	-

Appendix C. Parameters of the DFIG WTG Model

Table C.6
DFIG-based WTG Parameter.

Parameters	Magnitude	Parameters	Magnitude
Number of blade [int]	3.0	Pmax	1.0p.u
Blade length	75.0m	Gear box ratio [int -]	1/89
Power control time constant (Te)	0.01s	Number of poles (p)	4.0
Voltage control gain (kv)	10.0p.u	Inertia constants (Hm) [kWs/kVA]	3.0
Time constant (Tp)	3.0s	Magnetization reactance (Xm)	3.0
Pitch control gain (Kp)	10.0p.u	Rotor reactance (Xr)	0.08
Qmin	-0.7p.u	Rotor resistance (Rr)	0.01
Qmax	0.7p.u	Stator reactance (Xs)	0.10
Pmin	0.0p.u	Stator resistance (Rs)	0.01

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